

Story of Geometry

Part 1





About the Course

This course gives students a glimpse into how the ideas that guide our study of geometry were first discovered.



Placement & Combining Tips

This course is intended as a precursor to Part 2; however, the order may be reversed to facilitate combining students.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
7	Story of Geometry: Part 1 1 time/week 15 min	String, Straight-Edge, and Shadow The Student Bible Atlas, Revised Edition Historical Atlas of the World 6th Ed.

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Story of Geometry: Part 1				
		Story of Geometry: Part 1		



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Story of Geometry: Part 1



String, Straight-Edge, and Shadow



The Student Bible Atlas, Revised Edition



Historical Atlas of the World 6th Ed.



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

(No Quick Links Assigned)

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[or scan the QR](#)
[code for links.](#)



Story of Geometry: Part 1

How To Teach



Recap

Connect back to previous lesson.



Introduce

Optional: Use a picture, map, or engaging question to prepare the student to receive the idea from the day's lesson. Talk about one or two unknown words that will be crucial to understanding.



Read

Read the new passage out loud or have the student read it orally or silently.



Narrate

Have the student narrate orally or in writing.



Discuss

- Give students an opportunity to share their thoughts on the lesson and any connections made.



Practice

Optional: Attempt to recreate some of the problems from the stories.

Story of Geometry: Part 1

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Term 1

WEEK 1 15m Story of Geometry: Part 1 - Lesson 1

Introduction

Materials: String, Straight-Edge, and Shadow

→ INTRO

As you begin your study of geometry, this book will help you understand some concepts better by showing you how they were discovered and used to solve real problems.

→ READ, NARRATE, & DISCUSS

"Prologue" p.7-10 (UK 7-9) "The string, the" - "and the shadow."

WEEK 2 15m Story of Geometry: Part 1 - Lesson 2

The Natural Sense of Order

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.1 p.12-13 (UK 12-13) "Geometry itself" - "built into them."

- Look around the room in which you are sitting or outside. Write a paragraph telling what elements of mathematics (including geometry) you see.

WEEK 3 15m Story of Geometry: Part 1 - Lesson 3

Mathematics as Beauty

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.1 p.13-14 (UK 13-14) "Within ourselves" - "geometry really began."

WEEK 4 15m Story of Geometry: Part 1 - Lesson 4

From Fear to Wonder

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.2 p.15-16 (UK 15-16) "When and how did" - "an amazing surprise."

★ TEACHER NOTE

You may wish to preview p.15, which talks about the age of the earth. Also, please note the illustration of the partially nude woman on p.15 (UK 15).

Story of Geometry: Part 1

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Term 1

WEEK 5 15m Story of Geometry: Part 1 - Lesson 5

Structure in Nature

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.2 p.17-18 (UK 16-17) "Before we learn" - "of living things."

- How could seeing basic geometric shapes help you with nature study?
- Describe an object in terms of its shape. Use geometry words such as line, curve, spiral, perpendicular, etc.

WEEK 6 15m Story of Geometry: Part 1 - Lesson 6

Geometry of Living Things

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.2 p.18-19 (UK 17-18) "Did you ever" - "five-pointed star."

WEEK 7 15m Story of Geometry: Part 1 - Lesson 7

From the Art Gallery of the Universe

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.2 p.19-20 (UK 18-19) "Best of all" - "dim Stone Age."

WEEK 8 15m Story of Geometry: Part 1 - Lesson 8

Technics and Art

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.3 p.21-22 (UK 20-21) "Stone Age men" - "variety of nature's forms."

★ TEACHER NOTE

Please note the illustration of the partially nude woman on p.23 (UK 21).

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Term 1

WEEK 9 15m Story of Geometry: Part 1 - Lesson 9

Design Mimics Nature

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.3 p.22-24 (UK 21-22) "Perhaps unconsciously" - "using circles."

★ TEACHER NOTE

Please note the illustration of the partially nude woman on p.23 (UK 21).

WEEK 10 15m Story of Geometry: Part 1 - Lesson 10

Circles

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ INTRO

Where in this room do you see circles?

→ READ, NARRATE, & DISCUSS

Ch.3 p.24-25 (UK 22) "Today we are" - "they gave up!"

WEEK 11 15m Story of Geometry: Part 1 - Lesson 11

Drawing a Perfect Circle with String

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ INTRO

How would you define an "accurate" circle?

→ READ, NARRATE, & DISCUSS

Ch.3 p.26-27 (UK 22-24) "So, you see" - "the Stone Age."

WEEK 12 15m Story of Geometry: Part 1 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:

String, Straight-Edge, and Shadow

Story of Geometry: Part 1

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Term 2

WEEK 1 ☐ 15m Story of Geometry: Part 1 - Lesson 13

Agrarian Society

☐ Materials: String, Straight-Edge, and Shadow; Student Bible Atlas

→ INTRO

This section is titled "Reading the Shadows." What do you think that means?

→ READ, NARRATE, & DISCUSS

Ch.4 p.29-30 (UK 26-27) "Here our story" - "of the shadows."

- This map shows the Fertile Crescent. What do you notice about the surrounding land?

☐ Map Resource: Student Bible Atlas "Map 1: Lands of the Bible" p.4-5

WEEK 2 ☐ 15m Story of Geometry: Part 1 - Lesson 14

Noticing Shadows

☐ Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.4 p.30-32 (UK 27-29) "Perhaps it was" - "a repeated pattern."

WEEK 3 ☐ 15m Story of Geometry: Part 1 - Lesson 15

Measuring Time and Seasons

☐ Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.4 p.32-35 (UK 29-31) "The sun seemed" - "the fruitful year."

- Look at a sundial. How is it similar to, and different from, a clock?
- ∞ Image Link: Sundial

→ OBSERVE

Choose one of the ways to observe shadows from the book to try yourself.

WEEK 4 ☐ 15m Story of Geometry: Part 1 - Lesson 16

Usefulness of Right Angles

☐ Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

• IMPORTANT DATES

Village of Jarmo: First Evidence of Straight Lines & Right Angles (c.5000 BC, Syria)

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Term 2

→ INTRO

A surveyor is someone who makes very precise measurements of land to determine property boundaries. Why do you think that might be important?
Look at the picture of a modern surveyor. This story will show how this profession got its start.
∞ Image Link: Surveyor

→ READ, NARRATE, & DISCUSS

Ch.5 p.36-38 (UK 32-34) "Now at last" - "on the earth?"

WEEK 5 15m Story of Geometry: Part 1 - Lesson 17

The 3-4-5 Right Triangle

Materials: String, Straight-Edge, and Shadow

→ RECAP

What right angles did you notice last week?

→ READ, NARRATE, & DISCUSS

Ch.5 p.38-41 (UK 34-37) "We know part" - "perfect right angle."

→ PRACTICE

Using your geometry tools, try to construct a 3-4-5 right triangle.

• IMPORTANT DATES

Rope Stretchers (First Surveyors; c.3000 BC, Egypt)

WEEK 6 15m Story of Geometry: Part 1 - Lesson 18

The Plumb-Line

Materials: String, Straight-Edge, and Shadow

→ RECAP

Tell about a perfect right triangle.

→ READ, NARRATE, & DISCUSS

Ch.5 p.42-44 (UK 37-39) "Now that surveyors" - "way to keep peace."

- Where are levels and plumb lines important today?

WEEK 7 15m Story of Geometry: Part 1 - Lesson 19

Area and Architecture

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.5 p.44-46 (UK 39-41) "Again the idea" - "of the temple."

• IMPORTANT DATES

Oldest Man-Made Stone Structure (Step-Pyramid at Sakkara, c.2750 BC)
Great Pyramid (c.2850 BC)

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Term 2

WEEK 8 15m Story of Geometry: Part 1 - Lesson 20

Early Astronomy

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.6 p.47-50 (UK 42-44) "Meantime on the" - "along the Nile."

WEEK 9 15m Story of Geometry: Part 1 - Lesson 21

Arcs

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.6 p.50-52 (UK 44-46) "The Egyptians" - "exactly six equal parts."

→ PRACTICE

Using your geometry tools, try to construct a circle and divide it into six equal parts. Use the following pictures to help you.

Image Resource: String, Straight-Edge, and Shadow p.52-53 (UK 46-47)

WEEK 10 15m Story of Geometry: Part 1 - Lesson 22

Measuring Star Travel

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.6 p.52-54 (UK 46-48) "Anyhow, by chance" - "celestial bodies."

→ PRACTICE

Into how many equal parts is the circle of a clock divided by the hour marks? Using the same method you used last week to construct a circle with equal parts, construct a clock face (hours only).

• IMPORTANT DATES

First Continuous Astronomical Data Started (747 BC, Mesopotamia)

WEEK 11 15m Story of Geometry: Part 1 - Lesson 23

Circles as Instruments of Measure

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

Story of Geometry: Part 1

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Term 2	
→ INTRO Besides astronomy, in which fields of study or in what other endeavors do you think the circle could be used? → READ, NARRATE, & DISCUSS Ch.6 p.54-57 (UK 48-51) "Apart from" - "Near Eastern geometry."	
WEEK 12  15m Story of Geometry: Part 1 - Lesson 24 <i>Exam Week</i> → EXAMS Answer question(s) related to course. Questions will come from: String, Straight-Edge, and Shadow	

Story of Geometry: Part 1

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Term 3

WEEK 1 15m Story of Geometry: Part 1 - Lesson 25

Babylonians & Greeks

Materials: String, Straight-Edge, and Shadow; Student Bible Atlas

→ RECAP

What do you remember from last term?

→ READ, NARRATE, & DISCUSS

Ch.7 p.59-60 (UK 54-55) "At the start" - "the Black Sea."

→ LOCATE

Map Resource: Student Bible Atlas "Map 13: Babylon About 550 B.C." p.16
Look at the map of Babylon. Locate the Mediterranean and Black Seas (not labeled). Find Nineveh. What river runs by it? What Bible stories do you remember about Nineveh or King Nebuchadnezzar?

• IMPORTANT DATES

First Theoretical Geometry (c.6th century BC, Ionia)
Thales of Miletus (c.624 BC-c.548 BC)

WEEK 2 15m Story of Geometry: Part 1 - Lesson 26

From Practical to Theoretical Geometry

Materials: String, Straight-Edge, and Shadow; Historical Atlas of the World

→ RECAP

What did you read last time?

→ INTRO

Map Resource: Historical Atlas of the World "Classical Greece and Athenian Empire About 450 B.C." p.7
Find the colony of Ionia. Look to the south to find the city of Miletus. This is where today's story takes place.

→ READ, NARRATE, & DISCUSS

Ch.7 p.60-61 (UK 55-57) "So the 6th century" - "first scientists."

- Tell how the location of Miletus influenced its knowledge and development.

WEEK 3 15m Story of Geometry: Part 1 - Lesson 27

Thales: Founder of Geometry

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.7 p.61-62 (UK 57) "The Greeks had" - "founder of geometry."

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Term 3

WEEK 4 15m Story of Geometry: Part 1 - Lesson 28

Thales and Scientific Inquiry

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.8 p.63-64 (UK 58-59) "Thales, the father" - "the next season."

WEEK 5 15m Story of Geometry: Part 1 - Lesson 29

Importance of Observation

Materials: String, Straight-Edge, and Shadow

→ RECAP

What was Thales studying in last week's lesson? What do you think he will do?

→ READ, NARRATE, & DISCUSS

Ch.8 p.65-67 (UK 59-61) "Now he made" - "under your feet."

- Tell one of the stories of Thales.

WEEK 6 15m Story of Geometry: Part 1 - Lesson 30

Practical + Theoretical = New Ideas

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.8 p.67-68 (UK 61-62) "Nobody likes" - "theoretical geometry."

WEEK 7 15m Story of Geometry: Part 1 - Lesson 31

Magic & Mathematics

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.9 p.69-70 (UK 63-64) "In the Land" - "was a magician."

→ DISCUSS

How do you think Thales calculated the height of the Great Pyramid? Find something tall outside (a tree, telephone pole, etc.). Can you devise a way you could figure out its height?

Story of Geometry: Part 1

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Term 3

WEEK 8 15m Story of Geometry: Part 1 - Lesson 32

Proportion

Materials: String, Straight-Edge, and Shadow

→ RECAP

Were you able to figure out how to calculate the height of the tall object you chose?

→ READ, NARRATE, & DISCUSS

Ch.9 p.70-72 (UK 64-66) "To be sure" - "abstract proportion."

WEEK 9 15m Story of Geometry: Part 1 - Lesson 33

Shadows & Right Triangles

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.9 p.73-75 (UK 66-67) "Where others saw" - "height of the pyramid."

WEEK 10 15m Story of Geometry: Part 1 - Lesson 34

Finding the Height of a Tall Object

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ INTRO

Today we can try Thales' feat ourselves.

→ READ, NARRATE, & DISCUSS

Ch.9 p.75-76 (UK 67-70) "Before you hear" - "height of the pyramid."

→ PRACTICE

Use shadows and proportions to calculate the height of a tall object.

- Explain how Thales calculated the height of the pyramid.

WEEK 11 15m Story of Geometry: Part 1 - Lesson 35

The Abstract Right Triangle

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

Story of Geometry: Part 1

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Term 3	
→ READ, NARRATE, & DISCUSS Ch.9 p.76-78 (UK 70-71) "When he asked" - "of reasoning."	
WEEK 12  15m Story of Geometry: Part 1 - Lesson 36 <i>Exam Week</i> → EXAMS Answer question(s) related to course. Questions will come from: String, Straight-Edge, and Shadow	

Story of Geometry: Part 1

Examination

- Term 1** · String, Straight-Edge, and Shadow: Tell what you know about geometry in nature. OR Tell how design mimics nature.
- Term 2** · String, Straight-Edge, and Shadow: Tell how geometry is useful in architecture. OR Tell how humans learned to measure the seasons.
- Term 3** · String, Straight-Edge, and Shadow: Draw and explain how Thales calculated the height of the Great Pyramid.